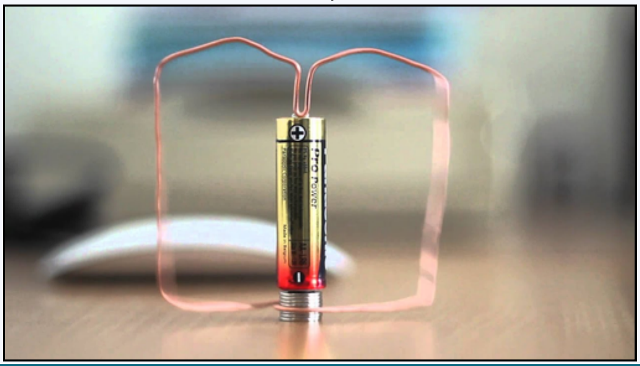
TECHNICAL DATA SHEET (FOR EXPERTS / TEACHERS) – WORKSHOP 3 : ELECTROMAGNETISM & FORCE/LAPLACE'S LAW

The homopolar motor



The Little Electric Train

<u>Background</u> : This educational workshop highlights the direct conversion of electrical energy into mechanical energy using permanent magnetic fields, without direct mechanical contact or gears.

<u>Objectives of the experiment</u> : To build the world's simplest electric train and a homopolar motor to demonstrate propulsion by axial and radial magnetic forces.

I. Physical Phenomena in Focus

- *<u>Lorentz Force and Laplace's Law</u>* : The train moves due to the interaction between an electric current **I** and a magnetic field **B**. When the battery and its magnets are inserted into the tunnel, the conductive magnets touch the bare copper wire, closing an electric circuit. A strong current **I** flows through the section of coils located between the two magnets. The elementary driving force (Laplace) acting on the train is defined by : $\boxed{F = \int I \cdot dl \times B}$
- *<u>Magnetic configuration (opposite poles)</u>* : To ensure that the forces add together vectorially rather than cancel each other out, the two neodymium magnets attached to the ends of the stack must be positioned with opposite magnetic phases (like poles facing outward).
- *<u>Homopolar motor</u>* : An axial permanent magnetic field (Wukong magnet at the base) intercepts a radial current flowing through the 1.3 mm tinned copper wire, generating a continuous torque that causes the wire to rotate in a stable mechanical motion.

II. Experimental Protocol and Technical Alignment

1. *<u>Solenoid fabrication</u>* : Wind the 1.3 mm copper wire onto the 16 mm PVC template in successive passes. The turns must be tight but not overlapping to avoid variations in inductance and mechanical friction.
2. *<u>Sliding contact calibration</u>* : The 15-mm-diameter magnets serve the dual purpose of current collectors and shaft guides. The 0.5-mm micro-clearance allows for a sliding electrical contact with low contact resistance (**Rc**).

DID YOU KNOW THAT ?

Did you know that ? How does the simplest train in the world work ? This little train doesn't need an internal combustion engine or gears. In fact, it uses pure electromagnetism ! As soon as you slide the battery into the copper tunnel, the magnets close an electrical circuit. A strong current flows through the copper and creates a magnetic field. Through the action-reaction effect (**Laplace's force**), the battery is literally propelled forward through the tube. This is the fundamental principle that powers all electric motors on Earth, from computer fans to modern electric cars and magnetic levitation (Maglev) trains ! *(Academic source: IEEE Transactions on Education).*